

DEPARTMENT
OF
ELECTRONICS & COMMUNICATION ENGINEERING

UG - B.E. COURSE
IN
Basic Electronics
Course Code : 21EC105 - 21EC205

I/ II Semester ODD & Even Semester

CLASS NOTES – Module 4

Academic Year : 2021-22

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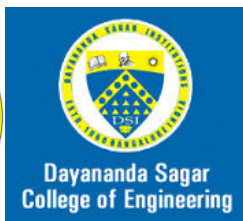
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**DEPARTMENT
OF
ELECTRONICS & COMMUNICATION ENGINEERING**

**Introduction to
Electronics
Engineering**

(Theory Notes)

Autonomous Course

Module – 4 Contents

Communication Systems 1 : Introduction, Elements of Communication Systems, Concept of modulation, methods of modulation - principles of AM, FM, AM / FM transmitters & receivers (block diagram description only), AM derivation.

Communication Systems 2: Block diagram and Principle of Optical Fiber Communication, Advantages and Applications of Optical Fiber communication.

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Block Diagram of Communication System

Figure shows the block diagram of a general communication system, in which the different functional elements are represented by blocks.

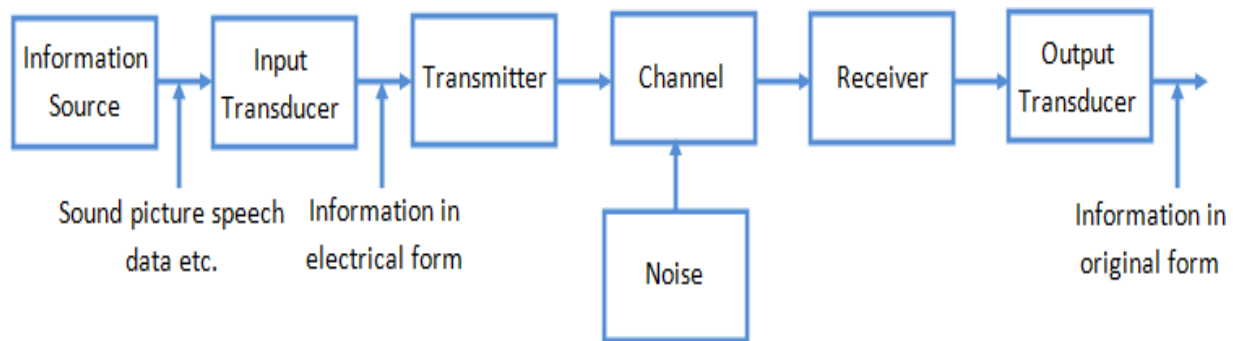


Figure1. Block diagram of Communication System.

The function of **information source** is to produce required message which has to be transmitted, for example text, voice, video etc.

A **transducer** is a device which converts one form of energy into another form.

The message from the information source may or may not be electrical in nature. In a case when the message produced by the information source is not electrical in nature, an input transducer is used to convert it into a time-varying electrical signal.

The function of the **transmitter** is to process the electrical signal from different aspects. Modulation is the main function of the transmitter. In modulation, the message signal is superimposed upon the high-frequency carrier signal. Other functions of transmitter are restriction of range of audio frequencies, amplification.

The term **channel** means the medium through which the message travels from the transmitter to the receiver.

The main function of the **receiver** is to reproduce the message signal in electrical form from the distorted received signal. This reproduction of the original signal is accomplished by a process known as the demodulation or detection. Demodulation is the reverse process of modulation carried out in transmitter.

Output transducer is the final stage which is used to convert an electrical message signal into its original form.

Modulation

In the modulation process, two signals are used namely the modulating signal and the carrier.

The modulating signal is nothing but the baseband signal or information signal while the carrier is a high frequency sinusoidal signal.

In the modulation process, some parameter of the carrier wave (such as amplitude, frequency or phase) is varied in accordance with the modulating signal. This modulated signal is then transmitted by the transmitter.

Need of Modulation

1. To Reduction in the height of antenna

For the transmission of radio signals, the antenna height must be multiple of $\lambda/4$, where λ is the wavelength.

$$\lambda = c / f$$

Where c: is the velocity of light, f: is the frequency of the signal to be transmitted

The minimum antenna height required to transmit a baseband signal of $f = 10$ kHz is calculated as follows:

$$\text{Minimum antenna height} = \frac{\lambda}{4} = \frac{c}{4f} = \frac{3 \times 10^8}{4 \times 10 \times 10^3} = 7500 \text{ meters i.e. } 7.5 \text{ km}$$

The antenna of this height is practically impossible to install.

Now, let us consider a modulated signal at $f = 1$ MHz. The minimum antenna height is given by,

$$\text{Minimum antenna height} = \frac{\lambda}{4} = \frac{c}{4f} = \frac{3 \times 10^8}{4 \times 10 \times 10^6} = 75 \text{ meters}$$

This antenna can be easily installed practically. Thus, modulation reduces the height of the antenna.

2. To Avoids mixing of signals

If the baseband sound signals are transmitted without using the modulation by more than one transmitter, then all the signals will be in the same frequency range i.e. 0 to 20 kHz. Therefore, all the signals get mixed together and a receiver cannot separate them from each other.

Hence, if each baseband sound signal is used to modulate a different carrier then they will occupy different slots in the frequency domain (different channels). Thus, modulation avoids mixing of signals.

3. To Increase the Range of Communication

The frequency of baseband signal is low, and the low frequency signals cannot travel long distance when they are transmitted. They get heavily attenuated.

The attenuation reduces with increase in frequency of the transmitted signal, and they travel longer distance.

The modulation process increases the frequency of the signal to be transmitted. Therefore, it increases the range of communication.

4. Multiplexing is possible

Multiplexing is a process in which two or more signals can be transmitted over the same communication channel simultaneously. This is possible only with modulation.

The multiplexing allows the same channel to be used by many signals. Hence, many TV channels can use the same frequency range, without getting mixed with each other or different frequency signals can be transmitted at the same time.

5. Improves Quality of Reception

With frequency modulation (FM) and the digital communication techniques such as PCM, the effect of noise is reduced to a great extent. This improves quality of reception.

Amplitude Modulation

Amplitude Modulation is defined as the amplitude (signal strength) of the carrier wave is varied in proportion to the waveform being transmitted.

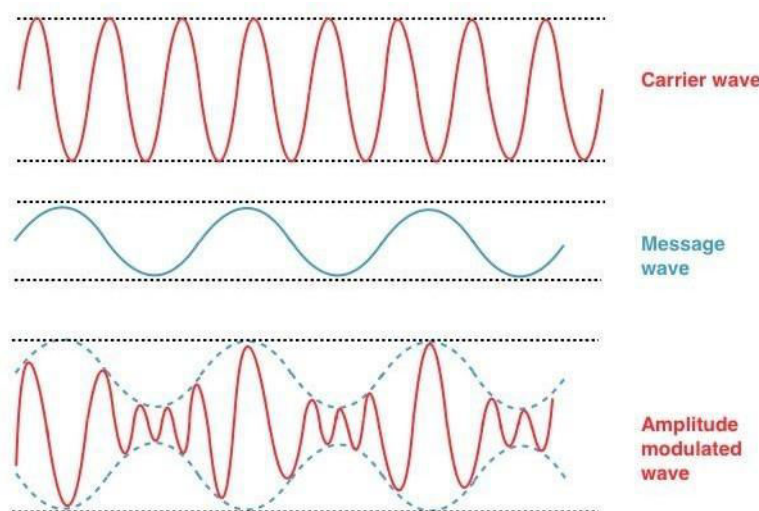


Figure 2. Amplitude modulated wave

In amplitude modulation, the amplitude of the carrier wave is varied in proportion to the instantaneous amplitude of the modulating signal, keeping its (carrier) frequency and phase constant. The modulating signal is given by

$$m(t) = V_m \sin(\omega_m t) \dots (1)$$

where,

V_m = Amplitude of the modulating signal

f_m = frequency of the modulating signal and

$$\omega_m = 2\pi f_m$$

The carrier signal is given by

$$c(t) = V_c \sin(\omega_c t) \dots (2)$$

where,

V_c = Amplitude of the carrier signal

f_c = frequency of the carrier signal and

$$\omega_c = 2\pi f_c$$

The amplitude of the AM signal is given by

$$V(t) = V_c + m(t) \dots (3)$$

Substitute Eq. (1) in Eq. (3), we get

$$V(t) = V_c + V_m \sin \omega_m t \dots (4)$$

The instantaneous voltage of AM wave is

$$V_{AM} = V(t) \sin \omega_c t \dots (5)$$

Substitute Eq. (4) in Eq. (5), we get

$$\begin{aligned} V_{AM} &= [V_c + V_m \sin \omega_m t] \sin \omega_c t \\ V_{AM} &= V_c \left[1 + \frac{V_m}{V_c} \sin \omega_m t \right] \sin \omega_c t \dots (6) \end{aligned}$$

where

$$\mu = m = \frac{V_m}{V_c}$$

$$\begin{aligned} V_{AM} &= V_c [1 + m \sin \omega_m t] \sin \omega_c t \\ V_{AM} &= V_c \sin \omega_c t + m V_c (\sin \omega_c t)(\sin \omega_m t) \dots (7) \end{aligned}$$

By using trigonometric relation,

$$\begin{aligned} \sin A \sin B &= \frac{1}{2} [\cos(A - B) - \cos(A + B)] \\ V_{AM} &= \underbrace{V_c \sin \omega_c t}_{\text{Carrier}} + \underbrace{\frac{m V_c}{2} \cos(\omega_c - \omega_m) t}_{\text{lower sideband}} - \underbrace{\frac{m V_c}{2} \cos(\omega_c + \omega_m) t}_{\text{upper sideband}} \dots (8) \end{aligned}$$

Eq(8) is the AM wave equation. It has 3 components.

- i) Unmodulated carrier wave
- ii) Lower side band (LSB) and
- iii) Upper side band (USB).

Frequency Spectrum

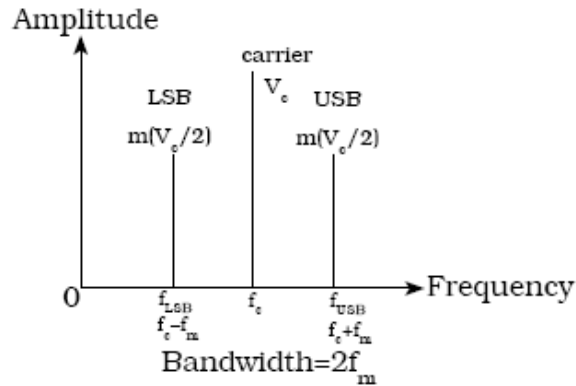


Figure 3. Frequency Spectrum

Bandwidth of AM wave is given by

$$\begin{aligned}
 \text{Bandwidth} &= f_{\text{USB}} - f_{\text{LSB}} \\
 &= f_c + f_m - (f_c - f_m) \\
 &= f_c + f_m - f_c + f_m
 \end{aligned}$$

$$\boxed{\text{Bandwidth} = 2 f_m}$$

Modulation Index:

Modulation index (**m** or μ) is the ratio of peak amplitude of the modulating signal to the peak amplitude of the carrier signal.

$$\boxed{m = \mu = \frac{V_m}{V_c}}$$

Modulation Index is expressed in percentage and is called percentage modulation

$$\boxed{\% m = \frac{V_m}{V_c} \times 100\%}$$

Note:

- 1) For proper amplitude modulation, A_m should be always be less than A_c .
- 2) Modulation index lies in between 0 and 1.

Frequency Modulation

Frequency Modulation is defined as the frequency of the carrier wave is varied in proportion to the waveform being transmitted.

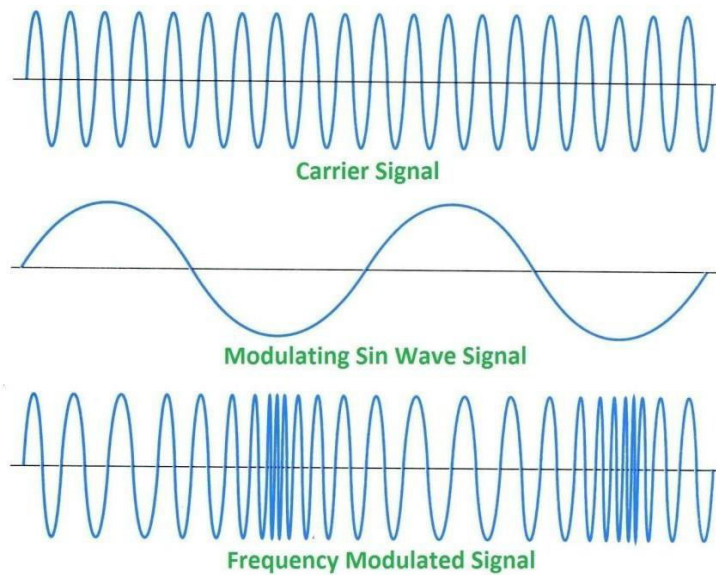


Figure 5. Frequency Modulation

AM/FM Transmitter Block Diagram

Below is an AM transmitter block diagram is simple to understand, the audio from microphone to antenna or aerial. Here is an explanation block diagram of AM transmitter:

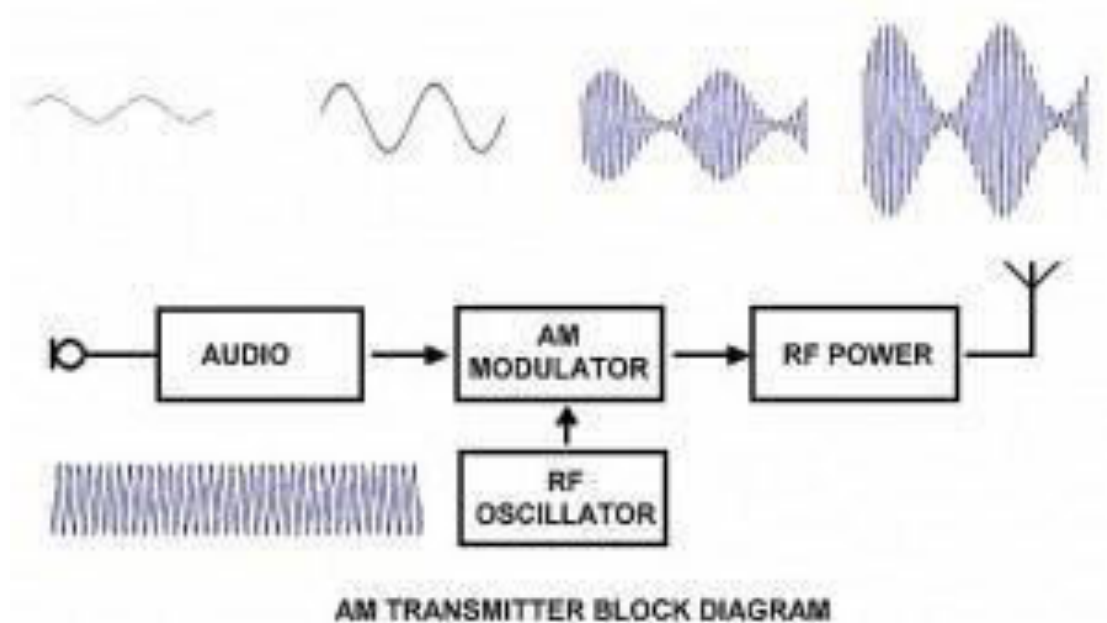


Figure 6. AM/FM Transmitter Circuit-Block Diagram

- **Audio:** an audio block is set regarding processing of audio and voice-related quality of audio. This usually consists of control channels, audio amplifiers, and audio processor (on a commercial press).
- **AM Modulator:** this block is where the modulation amplitude of the RF carrier from the oscillator does. Audio signal is modulated onto a radio frequency carrier to the modulation stage. By varying the modulation can be (height) the amplitude of the carrier is known as amplitude modulation (AM).
- **RF Oscillator:** here defined frequency or frequency range in which the transmitter will operate, in order to avoid interference to other frequencies or frequency bands. For commercial purposes, you must have a registered license.
- **RF Power r:** The signal amplitude modulation which has a deviation is permitted will be strengthened in this block. Of the power of reinforcement will be categorized from low to high power. The higher the power is emitted, the more stringent requirements of the RF filter. The next stage of the signal will be emitted through the correctly matched antenna to the entire coverage area.

AM/FM super heterodyne receiver

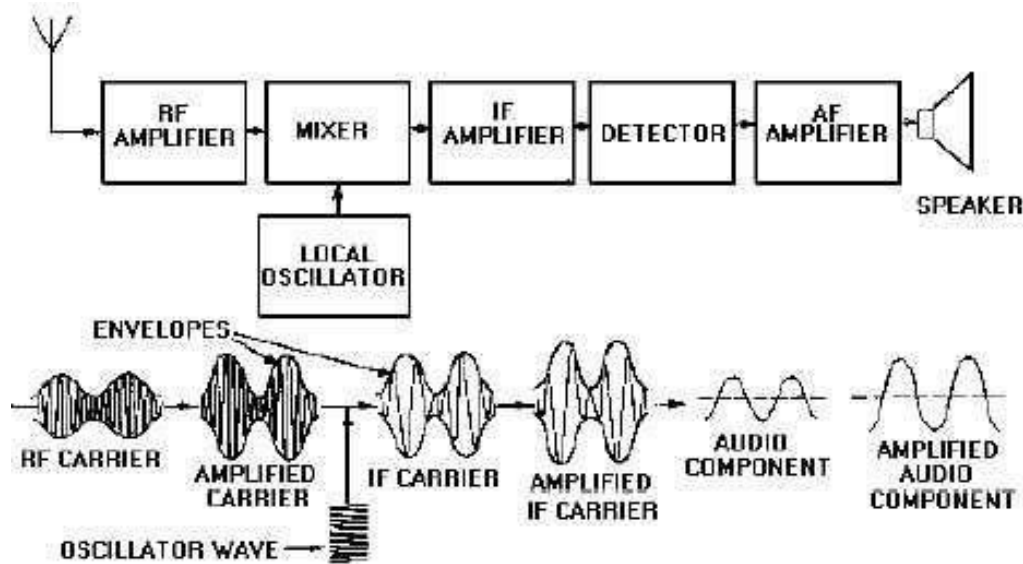


Figure 7. AM/FM super heterodyne receiver and waveforms.

The Superhet Receiver

The diagram shows a *super-sonic heterodyne* - or *superhet* - receiver, the standard pattern for receivers in general use today. The first thing to note is that three amplifiers are shown, the RF amplifier, the IF amplifier, and the AF amplifier.

The Radio Frequency Amplifier

This provides amplification for the signal as soon as it arrives from the antenna. The amplified signal is then passed to the *mixer/oscillator*. Mixers. The purpose of the mixer/oscillator is to translate the frequency of the incoming signal to the *intermediate frequency*, i.e. to the *IF amplifier*.

The mixer stage is usually acknowledged as being the noisiest stage in the receiver so an RF amplifier is positioned ahead of it to mask that noise with a higher signal level.

The RF amplifier stage should use a low-noise amplifying device - such as a low- noise transistor - to keep the internally- generated noise of the receiver to as low as possible. All the following amplifying stages will amplify this RF stage noise as well as the signal, so a low-noise device at the start of the receiving process is very important.

The Intermediate Frequency Amplifier

It is in the IF amplifier where most of the amplification in a receiver takes place.

From the IF stages, the signal passes to a detector. Here demodulation of the radio- frequency signal takes place to produce an audio signal.

The Audio Frequency Amplifier

Finally the audio signal is amplified in the audio amplifier and passed on to a speaker or phones for the listener.

Pulse Analog And Digital Modulation

Pulse modulation is further divided into analog and digital modulation.

The analog modulation techniques are mainly classified into

1. Pulse Amplitude Modulation,
2. Pulse Duration Modulation/Pulse Width Modulation, and
3. Pulse Position Modulation.

GENERAL OVERVIEW OF OPTICAL FIBER COMMUNICATION SYSTEM

Like all other communication system, the primary objective of optical fiber communication system also is to transfer the signal containing information (voice, data, and video) from the source to the destination. The general block diagram of optical fiber communication system is shown in the figure.

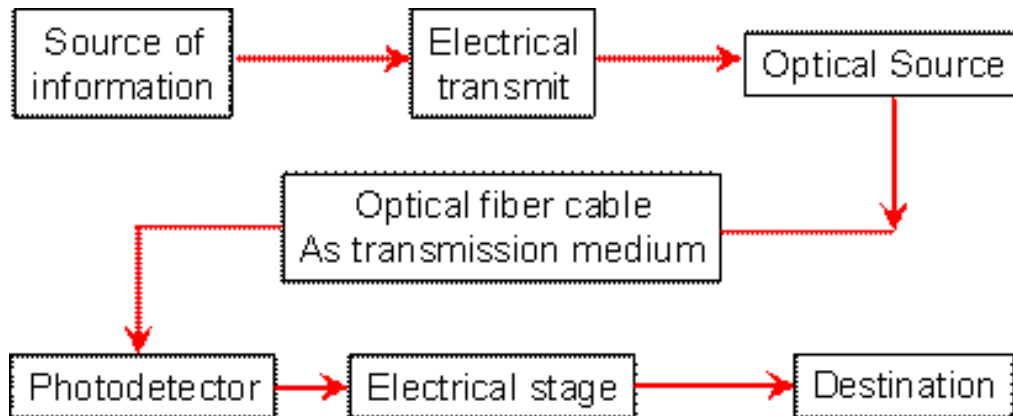


Figure11. Block Diagram of Optical Fiber Communication

The source provides information in the form of electrical signal to the transmitter.

The electrical stage of the transmitter drives an optical source to produce modulated light wave carrier. The main parts of the transmitter section are a source (either a **LED** or a **LASER**), efficient coupling means to couple the output power to the fiber, a modulation circuit and a level controller for LASERs.

The information carrying light wave then passes through the transmission medium i.e. optical fiber cables in this system.

Now it reaches to the receiver stage where the optical detector demodulates the optical carrier and gives an electrical output signal to the electrical stage. The common types of optical detectors used are photodiodes (p- i-n, avalanche), phototransistors, photoconductors etc.

Finally the electrical stage gets the real information back and gives it to the concerned destination.

Advantages of Optical Fiber Cable

- Bandwidth

Fiber optic cables have a much greater bandwidth than metal cables. The amount of information that can be transmitted per unit time of fiber over other transmission media is its most significant advantage.

- Low Power Loss

An optical fiber offers low power loss, which allows for longer transmission distances. In comparison to copper, in a network, the longest recommended copper distance is 100m while with fiber, it is 2km.

- Interference

Fiber optic cables are immune to electromagnetic interference. It can also be run in electrically noisy environments without concern as electrical noise will not affect fiber.

- Size

In comparison to copper, a fiber optic cable has nearly 4.5 times as much capacity as the wire cable has and a cross sectional area that is 30 times less.

- Weight

Fiber optic cables are much thinner and lighter than metal wires. They also occupy less space with cables of the same information capacity. Lighter weight makes fiber easier to install.

- Security

Optical fibers are difficult to tap. As they do not radiate electromagnetic energy, emissions cannot be intercepted. As physically tapping the fiber takes great skill to do undetected, fiber is the most secure medium available for carrying sensitive data.

- Flexibility

An optical fiber has greater tensile strength than copper or steel fibers of the same diameter. It is flexible, bends easily and resists most corrosive elements that attack copper cable.

- Cost

The raw materials for glass are plentiful, unlike copper. This means glass can be made more cheaply than copper.

Disadvantages of Optical Fiber Cable

- Difficult to Splice

The optical fibers are difficult to splice, and there are loss of the light in the fiber due to scattering. They have limited physical arc of cables. If you bend them too much, they will break.

- Expensive to Install

The optical fibers are more expensive to install, and they have to be installed by the specialists. They are not as robust as the wires. Special test equipment is often required to the optical fiber.

- Highly Susceptible

The fiber optic cable is a small and compact cable, and it is highly susceptible to becoming cut or damaged during installation or construction activities. The fiber optic cables can provide tremendous data transmission capabilities. So, when the fiber optic cabling is chosen as the transmission medium, it is necessary to address restoration, backup and survivability.

- Can't Be Curved

The transmission on the optical fiber requires repeating at distance intervals. The fibers can be broken or have transmission losses when wrapped around curves of only a few centimeters radius.

Applications

Due to its variety of advantages optical fiber communication system has a wide range of application in different fields namely :

- Public network field** which includes trunk networks, junction networks, local access networks, submerged systems, synchronous systems etc.
- Field of military applications,**
- Civil, consumer and industrial applications,**
- Field of computers,** which is the center of research right now.

Vision & Mission of the Institute

Vision:

- ❖ To impart quality technical education with a focus on Research and Innovation emphasizing on Development of Sustainable and Inclusive Technology for the benefit of society.

Mission:

- ❖ To provide an environment that enhances creativity and Innovation in pursuit of Excellence.
- ❖ To nurture teamwork in order to transform individuals as responsible leaders and entrepreneurs.
- ❖ To train the students to the changing technical scenario and make them to understand the importance of sustainable and inclusive technologies.

Vision & Mission of the Department

Vision :

- ❖ To achieve continuous improvement in quality technical education for global competence with focus on industry, societal needs, research and professional success.

Mission:

- ❖ Offering quality education in Electronics and Communication Engineering with effective teaching learning process in multidisciplinary environment.
- ❖ Training the students to take-up projects in emerging technologies and work with team spirit.
- ❖ To imbibe professional ethics, development of skills and research culture for better placement opportunities.

Program Education Objectives

After four years, the students will be

- PEO1** : ready to apply the state-of-art technology in industry and meeting the societal needs with knowledge of Electronics and Communication Engineering due to strong academic culture.
- PEO2** : competent in technical and soft skills to be employed with capability of working in multidisciplinary domains.
- PEO3** : professionals, capable of pursuing higher studies in technical, research or management programs.

Program Specific Outcomes

- PSO1** : Design, develop and integrate electronic circuits and systems using current practices and standards.
- PSO2** : Apply the knowledge of hardware and software in designing Embedded & Communication systems.



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